

Project:	ETAP	Page:	1
Location:	20.6.0C	Date:	30-03-2026
Contract:		SN:	
Engineer:		Revision:	Base
Filename:	Belalang Hijau BETA	Config.:	Normal
	Study Case: LF		

Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (1): Design
Generation Category (1): Design
Load Diversity Factor: None

	Swing	V-Control	Load	Total
Number of Buses:	1	0	14	15

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	12	0	1	14

Method of Solution:	Adaptive Newton-Raphson Method
Maximum No. of Iteration:	99
Precision of Solution:	0.0001000
System Frequency:	50.00 Hz
Unit System:	Metric
Project Filename:	Belalang Hijau BETA
Output Filename:	C:\ETAP 2060\Belalang Hijau BETA16\Belalang Hijau BETA\Untitled.lfr

Project: **ETAP**
Location: **20.6.0C**
Contract:
Engineer:
Filename: Belalang Hijau BETA
Study Case: LF

Page: 2
Date: 30-03-2026
SN:
Revision: Base
Config.: Normal

Adjustments

<u>Tolerance</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Percent</u>
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		
<u>Temperature Correction</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Degree C</u>
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	kW	kvar	kW	kvar	kW	kvar	kW	kvar
Bus AUX	0.400	1	100.0	0.0			2.0	1.0				
Bus Crane	0.400	1	100.0	0.0	109.7	48.4						
Bus Gel Breaking	0.400	1	100.0	0.0	129.6	56.8						
Bus GTG	6.000	1	100.0	0.0								
Bus Heater	0.400	1	100.0	0.0			50.0	24.2				
Bus HT	0.400	1	100.0	0.0			67.5	32.7				
Bus Lighting P.	0.400	1	100.0	0.0			45.0	21.8				
Bus Mkomp	6.000	1	100.0	0.0	319.5	156.7						
Bus P. Trafo	6.000	1	100.0	0.0								
Bus S. Trafo	0.400	1	100.0	0.0								
Bus Sea Water	0.400	1	100.0	0.0	14.4	8.0						
Bus UPS	0.400	1	100.0	0.0			12.0	5.8				
Gen Bus Power Supply	6.000	1	100.0	0.0								
LV SW2	0.400	1	100.0	0.0								
LV Switchrack	0.400	1	100.0	0.0								
Total Number of Buses: 15					573.236	269.851	176.499	85.482	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			kvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	kW	kvar	% PF	Max	Min
Bus GTG	6.000	Swing	1	100.0	0.0					
						0.000	0.000			

Project:	ETAP	Page:	4
Location:	20.6.0C	Date:	30-03-2026
Contract:		SN:	
Engineer:	Study Case: LF	Revision:	Base
Filename:	Belalang Hijau BETA	Config.:	Normal

Line/Cable/Busway Input Data

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)

Line/Cable/Busway		Length							
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable AUX SP	0,6NCUS3	1.5	100.0	0.0	1	75	14.688140	0.100000	0.0000208
Cable Kompessor	10NCUS3	185	150.0	0.0	1	75	0.122083	0.087300	
Cable Motor Gel breaking	1,0NCUS3	120	30.0	0.0	1	75	0.186940	0.070500	
Cable1	1,0NCUS3	35	70.0	0.0	1	75	0.638076	0.074300	
Cable2	1,0NCUS3	25	70.0	0.0	1	75	0.884149	0.076800	
Cable3	1,0NCUS3	120	90.0	0.0	1	75	0.186940	0.070500	
Cable4	1,0NCUS3	25	50.0	0.0	1	75	0.884149	0.076800	
Cable5	1,0NCUS3	25	30.0	0.0	1	75	0.884149	0.076800	
Cable6	1,0NCUS3	25	50.0	0.0	1	75	0.884149	0.076800	
Cable8	1,0NCUS3	400	30.0	0.0	1	75	0.061042	0.069400	
Cable9	10NCUS3	25	30.0	0.0	1	75	0.884149	0.117000	
Cable11	10NCUS3	95	500.0	0.0	1	75	0.235582	0.095500	

Line / Cable / Busway resistances are listed at the specified temperatures.

Project:

Location:

Contract:

Engineer:

Filename:

ETAP

20.6.0C

Study Case: LF

Belalang Hijau BETA

Page: 5

Date: 30-03-2026

SN:

Revision: Base

Config.: Normal

2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	kVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
Power Trafo	3-Phase	500.0	6.000	0.400	4.00	1.50	0	0	0	-2.500	0	4.0000	Dyn	0.000

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
Power Trafo	2W XFMR	Bus P. Trafo	Bus S. Trafo	432.67	649.00	780.00	
Cable AUX SP	Cable	LV SW2	Bus AUX	91800.84	625.00	91802.97	0.0000003
Cable Kompessor	Cable	Gen Bus Power Supply	Bus Mkomp	5.09	3.64	6.25	
Cable Motor Gel breaking	Cable	LV SW2	Bus Gel Breaking	350.51	132.19	374.61	
Cable1	Cable	LV Switchrack	Bus Heater	2791.58	325.06	2810.44	
Cable2	Cable	LV Switchrack	Bus Lighting P.	3868.15	336.00	3882.72	
Cable3	Cable	LV Switchrack	Bus Crane	1051.54	396.56	1123.83	
Cable4	Cable	LV Switchrack	Bus Sea Water	2762.97	240.00	2773.37	
Cable5	Cable	LV Switchrack	Bus HT	1657.78	144.00	1664.02	
Cable6	Cable	LV Switchrack	Bus UPS	2762.97	240.00	2773.37	
Cable8	Cable	Bus S. Trafo	LV Switchrack	114.45	130.13	173.30	
Cable9	Cable	Gen Bus Power Supply	Bus P. Trafo	7.37	0.98	7.43	
Cable11	Cable	Bus GTG	Gen Bus Power Supply	32.72	13.26	35.31	
CB21	Tie Breakr	LV Switchrack	LV SW2				

Project:	ETAP	Page:	7
Location:	20.6.0C	Date:	30-03-2026
Contract:		SN:	
Engineer:		Revision:	Base
Filename:	Belalang Hijau BETA	Config.:	Normal

Study Case: LF

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow				XFMR	
ID	kV	% Mag.	Ang.	kW	kvar	kW	kvar	ID	kW	kvar	Amp	%PF	%Tap
Bus AUX	0.400	96.430	-0.8	0.0	0.0	1.9	0.9	LV SW2	-1.858	-0.900	3.1	90.0	
Bus Crane	0.400	96.818	-1.2	0.0	0.0	109.7	48.4	LV Switchrack	-109.702	-48.372	178.7	91.5	
Bus Gel Breaking	0.400	97.666	-1.3	0.0	0.0	129.6	56.8	LV SW2	-129.594	-56.836	209.1	91.6	
* Bus GTG	6.000	100.000	0.0	759.2	370.2	0.0	0.0	Gen Bus Power Supply	759.189	370.231	81.3	89.9	
Bus Heater	0.400	96.780	-1.0	0.0	0.0	46.8	22.7	LV Switchrack	-46.835	-22.683	77.6	90.0	
Bus HT	0.400	97.075	-1.0	0.0	0.0	63.6	30.8	LV Switchrack	-63.609	-30.807	105.1	90.0	
Bus Lighting P.	0.400	96.456	-0.9	0.0	0.0	41.9	20.3	LV Switchrack	-41.867	-20.277	69.6	90.0	
Bus Mkomp	6.000	99.680	0.0	0.0	0.0	319.5	156.7	Gen Bus Power Supply	-319.547	-156.669	34.4	89.8	
Bus P. Trafo	6.000	99.668	0.0	0.0	0.0	0.0	0.0	Gen Bus Power Supply	-437.068	-212.546	46.9	89.9	
								Bus S. Trafo	437.068	212.546	46.9	89.9	-2.500
Bus S. Trafo	0.400	98.961	-1.1	0.0	0.0	0.0	0.0	LV Switchrack	427.037	197.501	686.2	90.8	
								Bus P. Trafo	-427.037	-197.500	686.2	90.8	
Bus Sea Water	0.400	97.782	-1.2	0.0	0.0	14.4	8.0	LV Switchrack	-14.392	-7.975	24.3	87.5	
Bus UPS	0.400	97.870	-1.2	0.0	0.0	11.5	5.6	LV Switchrack	-11.491	-5.566	18.8	90.0	
Gen Bus Power Supply	6.000	99.702	0.0	0.0	0.0	0.0	0.0	Bus Mkomp	319.612	156.716	34.4	89.8	
								Bus P. Trafo	437.243	212.569	46.9	89.9	
								Bus GTG	-756.855	-369.285	81.3	89.9	
LV SW2	0.400	98.208	-1.3	0.0	0.0	0.0	0.0	Bus AUX	1.900	0.900	3.1	90.4	
								Bus Gel Breaking	130.330	57.113	209.1	91.6	
								LV Switchrack	-132.230	-58.013	212.2	91.6	
LV Switchrack	0.400	98.208	-1.3	0.0	0.0	0.0	0.0	Bus Heater	47.642	22.777	77.6	90.2	
								Bus Lighting P.	42.767	20.355	69.6	90.3	
								Bus Crane	111.315	48.980	178.7	91.5	
								Bus Sea Water	14.470	7.982	24.3	87.6	
								Bus HT	64.488	30.884	105.1	90.2	
								Bus UPS	11.538	5.570	18.8	90.1	
								Bus S. Trafo	-424.450	-194.560	686.2	90.9	
								LV SW2	132.230	58.013	212.2	91.6	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic		kVA	% PF	Amp	Percent Loading
ID	kV	Rated Amp	kW	kvar	kW	kvar	kW	kvar	kW	kvar				
Bus AUX	0.400				1.858	0.900					2.064	90.0	3.1	
Bus Crane	0.400		109.702	48.372							119.893	91.5	178.7	
Bus Gel Breaking	0.400		129.594	56.836							141.509	91.6	209.1	
Bus GTG	6.000										844.653	89.9	81.3	
Bus Heater	0.400		0.000	0.000	46.835	22.683					52.039	90.0	77.6	
Bus HT	0.400		0.000	0.000	63.609	30.807					70.677	90.0	105.1	
Bus Lighting P.	0.400		0.000	0.000	41.867	20.277					46.519	90.0	69.6	
Bus Mkomp	6.000		319.547	156.669							355.887	89.8	34.4	
Bus P. Trafo	6.000		0.000	-0.001							486.008	89.9	46.9	
Bus S. Trafo	0.400		0.000	-0.001							470.496	90.8	686.2	
Bus Sea Water	0.400		14.392	7.975							16.454	87.5	24.3	
Bus UPS	0.400		0.000	0.000	11.491	5.566					12.768	90.0	18.8	
Gen Bus Power Supply	6.000			0.000							842.140	89.9	81.3	
LV SW2	0.400										144.396	91.6	212.2	
LV Switchrack	0.400										466.917	90.9	686.2	

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).

Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (kVA)	Loading (input)		Loading (output)	
						kVA	%	kVA	%
Cable AUX SP	Cable	20.49	3.09	15.08					
Cable Motor Gel breaking	Cable	299.85	209.13	69.75					
Cable1	Cable	136.89	77.61	56.70					
Cable2	Cable	110.81	69.61	62.82					
Cable3	Cable	299.85	178.74	59.61					
Cable4	Cable	110.81	24.29	21.92					
Cable5	Cable	110.81	105.09	94.83					
Cable6	Cable	110.81	18.83	16.99					
Power Trafo	Transformer				500.0	486.0	97.2	470.5	94.1

* Indicates a branch with operating load exceeding the branch capability.

Project:	ETAP	Page:	10
Location:	20.6.0C	Date:	30-03-2026
Contract:		SN:	
Engineer:	Study Case: LF	Revision:	Base
Filename:	Belalang Hijau BETA	Config.:	Normal

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	kw	kvar	kw	kvar	kW	kvar	From	To	
Cable AUX SP	-1.9	-0.9	1.9	0.9	0.0	0.0	96.4	98.2	1.78
Cable Kompessor	-319.5	-156.7	319.6	156.7	0.1	0.0	99.7	99.7	0.02
Cable Motor Gel breaking	-129.6	-56.8	130.3	57.1	0.7	0.3	97.7	98.2	0.54
Cable1	-46.8	-22.7	47.6	22.8	0.8	0.1	96.8	98.2	1.43
Cable11	759.2	370.2	-756.9	-369.3	2.3	0.9	100.0	99.7	0.30
Cable2	-41.9	-20.3	42.8	20.4	0.9	0.1	96.5	98.2	1.75
Cable3	-109.7	-48.4	111.3	49.0	1.6	0.6	96.8	98.2	1.39
Cable4	-14.4	-8.0	14.5	8.0	0.1	0.0	97.8	98.2	0.43
Cable5	-63.6	-30.8	64.5	30.9	0.9	0.1	97.1	98.2	1.13
Cable6	-11.5	-5.6	11.5	5.6	0.0	0.0	97.9	98.2	0.34
Cable8	427.0	197.5	-424.5	-194.6	2.6	2.9	99.0	98.2	0.75
Cable9	-437.1	-212.5	437.2	212.6	0.2	0.0	99.7	99.7	0.03
Power Trafo	437.1	212.5	-427.0	-197.5	10.0	15.0	99.7	99.0	0.71
					20.3	20.1			

* This Transmission Line includes Series Capacitor.

Project: **ETAP**
Location: **20.6.0C**
Contract:
Engineer:
Filename: Belalang Hijau BETA
Study Case: LF

Page: 11
Date: 30-03-2026
SN:
Revision: Base
Config.: Normal

Alert Summary Report

% Alert Settings

Loading

	<u>Critical</u>
Bus	100.0
Cable / Busway	100.0
Reactor	100.0
Line	100.0
Transformer	100.0
Panel	100.0
Protective Device	100.0
Generator	100.0
Inverter/Charger	100.0

Bus Voltage

OverVoltage	105.0
UnderVoltage	95.0

Generator Excitation

OverExcited (Q Max.)	100.0
UnderExcited (Q Min.)	100.0

Project:	ETAP	Page:	12
Location:	20.6.0C	Date:	30-03-2026
Contract:		SN:	
Engineer:	Study Case: LF	Revision:	Base
Filename:	Belalang Hijau BETA	Config.:	Normal

SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	kW	kvar	kVA	% PF
Source (Swing Buses):	759.2	370.2	844.7	89.88 Lagging
Source (Non-Swing Buses):	0.0	0.0	0.0	
Total Demand:	759.2	370.2	844.7	89.88 Lagging
Total Motor Load:	573.2	269.9	633.6	90.48 Lagging
Total Static Load:	165.7	80.2	184.1	90.00 Lagging
Total Constant I Load:	0.0	0.0	0.0	
Total Generic Load:	0.0	0.0	0.0	
Apparent Losses:	20.3	20.1		
System Mismatch:	0.0	0.0		

Number of Iterations: 2